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Pupil size changes reveal that dogs are sensitive to the social conventions behind human gestures

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ABSTRACT

While many animals display behaviours resembling social conventions when interacting with conspecifics, it is unknown whether animals who live in close contact with humans develop sensitivity to our social conventions. With an eye-tracking study ($N = 39$), we asked whether dogs' (*Canis familiaris*) eye movements and pupil sizes would show a reaction when observing a violation in the customary response to human conventional communicative gestures ("come here" and "go away"). In the video stimuli, a human actress either approached or walked away from a second human actress. Both these actions (approaching or walking away) could be expected or surprising (within-subject), depending on whether they happened in response to the "come here" or the "go away" gesture of the second actress. While dogs' looking times and number of gaze transitions between the two actresses did not differ significantly between congruent and incongruent responses, their pupil size changes did: dogs' pupils dilated more when watching the incongruent responses (approaching after the "go away" gesture and walking away after the "come here" gesture) compared to the congruent ones. We discuss the possibility that pet dogs, during their ontogeny, implicitly form expectations about the usual response to some conventional human gestures. These expectations are not self-centred, as they were evidenced when dogs observed a third-party interaction between two humans, and they are context-specific, as the same responses elicited surprise following one gesture but not the other. We also consider alternative explanations and limitations of our findings.

Social conventions, descriptive regularities about the usual behaviour in specific circumstances, are pervasive in human societies (Heyes, 2024; Schmidt & Rakoczy, 2023). In particular, some intransitive gestures in some cultures are well understood as communicative conventions. For example, if we observe someone raise a hand and wave at another person, we immediately and effortlessly infer the intention to "greet". Human expectations about the conventions in third-party social interactions begin to develop during the second half of the first year of life (Beier & Spelke, 2012). For example, 7- to 12-month olds have expectations about how members of the same social group should act (Powell & Spelke, 2013, 2018) and 11-month olds' pupil size changes show surprise when conformity does not lead to social inclusion and when non-conformity leads to social inclusion (Köster & Hepach, 2024). By their first year, human infants already expect language and artifacts to be used in a conventional way (Schmidt & Rakoczy, 2023). However, we still do not know the extent to which the ability to detect violations of social conventions, from the observation of non-verbal third-party interactions, is shared with other animals.

Dogs (*Canis familiaris*) are thought to follow some forms of social conventions during social play (Bekoff, 2015) and constitute a promising model for the study of the development and evolution of social convention understanding. Specifically, they have been proposed as a model for the study of socio-cognitive processes that might be functionally analogous to those of humans as a result of convergent evolution, due to our shared social environment (Hare & Tomasello, 2005; Kaminski & Marshall-Pescini, 2014; Topál, Miklósi, et al., 2009). For example, dogs are excellent readers of human non-verbal communication, to which they often respond appropriately (for reviews, see Huber, 2016; Huber & Lonardo, 2023; Kaminski & Nitzschner, 2013). This includes an understanding of pointing gestures both as referential (Völter et al., 2025) and as indicating a location (Lakatos et al., 2012; Tauzin et al., 2015). However, dogs do not follow all types of pointing gestures equally well: they have difficulty with gestures which are less visually clear or less commonly encountered, such as elbow cross-pointing,

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forward cross-pointing and knee-pointing, especially when the human body parts conveying the pointing signal do not protrude from the pointer's body (Soproni et al., 2001). In contrast, they succeed with more direct or familiar gestures, such as arm or leg pointing (Lakatos et al., 2009). Concerning the precise strategy used to follow a pointing gesture to its possible referent, dogs are unlikely to follow the precise linear trajectory along pointing arms. Instead, they seem to follow the side of the human gesturing because, when multiple possible referents are on the same side of the pointer, they rely on heuristics such as picking the referent closer to the pointer (Lakatos et al., 2012). Further, their point following to locate hidden food might not be as robust as previously thought; when showing a contralateral momentary (2 s) pointing gesture, dogs ($N = 455$) performed only minimally above chance level, irrespective of getting ostensive cues before (mean = 53% correct) or not (mean = 52% correct) in a two-choice task (ManyDogs Project, et al., 2023). While the familiarity of the pointer does not affect performance, sometimes dogs struggle to ignore misleading human communicative gestures, especially if these are preceded by ostensive cues (Barnard et al., 2019; Chijiiwa et al., 2021; Kundey et al., 2010; Marshall-Pescini et al., 2012; Petter et al., 2009; Szetei et al., 2003; Topál, Gergely, et al., 2009). In addition to pointing, during their ontogeny in human households, pet dogs, like children, are extensively exposed to human social conventions. Sometimes they might even be explicitly trained to respect some conventions (e.g., approach a human who taps with her hand a certain surface). Specifically, dogs are routinely trained to associate human hand gestures (commands) with actions. Studies with rescue dogs have shown the primary role of gestural over verbal communication when humans are asking dogs to obey commands (D'Aniello et al., 2016). Moreover, once learned the gestural signals (e.g., for "sit", "down", "stay" and "come"), rescue dogs respond to them irrespective of whether the gesture is emitted by their owner or a stranger (Scandurra et al., 2017). However, in previous behavioural studies, unlike human children, dogs did not seem to follow a human experimenter's deictic gesture when this was directed to another human (Kaminski et al., 2012) nor did they evaluate a human experimenter based on her interaction with another dog (Nitzschner et al., 2012). Hence, it is not known whether dogs have expectations about the usual responses to human non-verbal conventional gestures, when they observe interacting humans.

To investigate dogs' expectations about third-party non-verbal communicative gestures, we showed them video recordings of social interactions between two human actresses, one of which (the signaller) communicated to the other (the receiver) to approach her or to move away from her (Fig. 1). For the "come here" signal the signaller showed a series of conventional gestures associated with the intention to invite the receiver to approach (Guidetti & Nicoladis, 2008). These gestures are

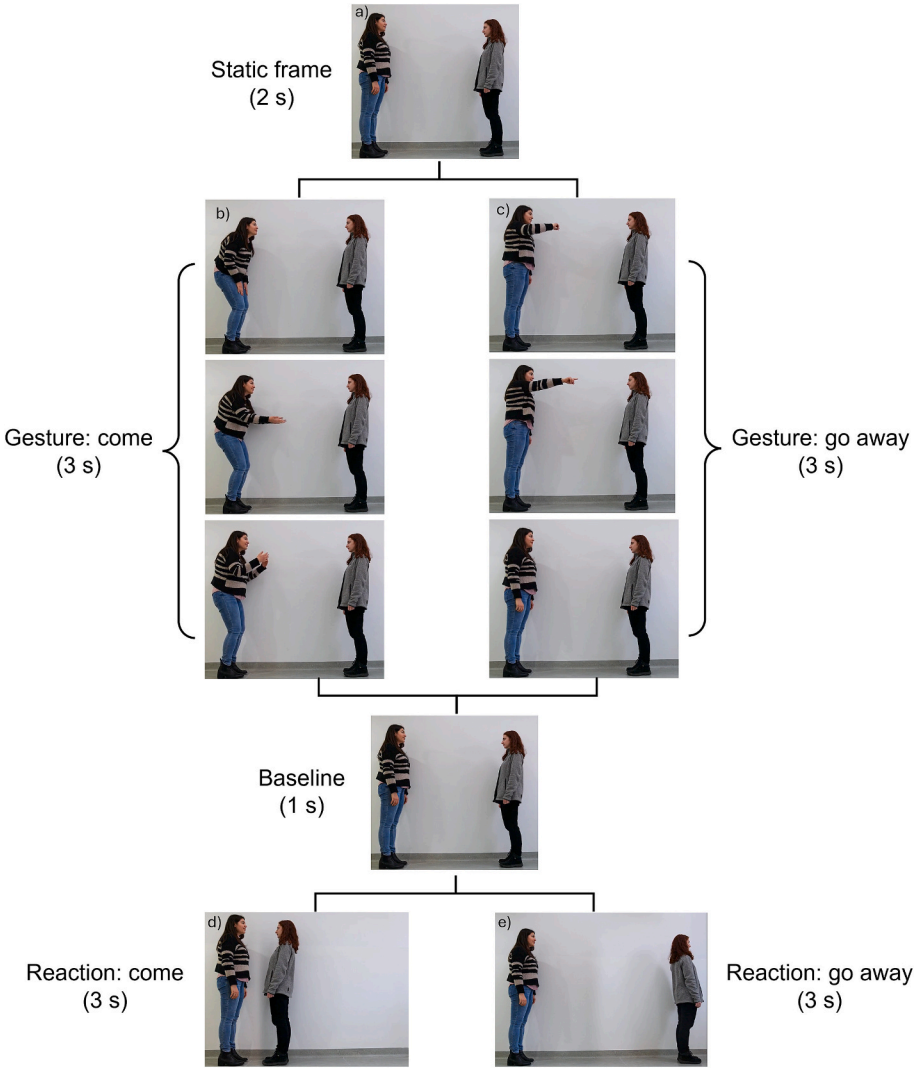


Fig. 1. The sequence of events shown in the experiment: *(a)* Beginning of all sequences; *(b)* "come" gesture and *(c)* "go away" gestures; *(d)-(e)* after either of these gestures, dogs were presented with congruent or incongruent reactions (within-subject design).

even shared with nonhuman apes (Genty & Zuberbühler, 2015; Hobaiter & Byrne, 2014). Specifically, the signaller nodded her head with a positive facial expression, slightly crouched down bending her chest forward, patted her hands on her thighs and beckoned, to invite the other person to approach her. For the "go away" signal, the signaller extended her arm and index finger away from, and perpendicular to, her body, maintaining an upright posture and a serious facial expression, to communicate to the other person to move away from her. Deictic gestures are routinely used by humans to attract attention to a location distant from the signaller and they develop long before language does (Cartmill et al., 2011).

In a 2 × 2 within-subject design, we varied the sender's signal ("come here" or "go away") and whether the recipient's behavioural response was congruent or incongruent with the sender's signal. Following the "come here" signal, the congruent response was approaching the signaller, while the incongruent response was walking away from the signaller. In the "go away" condition, the congruent response was walking away, while the incongruent response was approaching. We hypothesised that, due to the experience they gain during their ontogeny in a human household (including their training), dogs acquired expectations about the customary behaviour the receiver will adopt in response to the two communicative cues. Hence, we predicted that seeing the congruent response in both conditions should result in shorter looking times to the actresses as compared to seeing the incongruent response, which should instead violate dogs' expectations and result in longer looking times when dogs witness the incongruent reactions. Importantly, because the same behaviour (approaching or walking away) is in one condition the customary (expected) reaction and in the other condition the incongruent (surprising) reaction, this design allows to test whether the observers' expectations are context-relative, a signature of human understanding of conventions (Rakoczy & Schmidt, 2013). Assuming that human communicative gestures can be considered social conventions (Lewis, 1969; Tomasello, 2007), we analysed the participants' pupil sizes as a measure of an orienting response and arousal elicited by a violation of a social convention. Our prediction was that their pupils would dilate more after observing the receiving actress's gesture-incongruent reaction than after observing a gesture-congruent reaction.

1. Methods

This experiment was discussed and approved by the ethics committee of the University of Veterinary Medicine of Vienna (ETK-195/12/2022; ETK-53/03/2023). Data were collected using an Eyelink 1000 eye-tracker (SR Research, Canada; sampling rate: 1000 Hz, monocular tracking of the right eye). Before taking part in the experiments, dogs underwent training (Karl et al., 2020) to lay their head on a chinrest (height adjusted depending on the dog's size), 70 cm away from the screen. Videos were presented on a 24-in. LCD monitor with resolution 1024 × 768 px and refresh rate 100 Hz, to accommodate dogs' flicker-fusion threshold (Coile et al., 1989). The criterion for successful training completion, and start of the tests, was that dogs performed a validation of a calibration with less than 1° of visual angle deviation and were able to maintain their head on the chinrest (without leaving when the trainer left their field of view) for up to one minute.

On testing days, each session began with a five-point calibration of each dog's gaze positions on the screen. We used animations (inside squares with a side length of 32 or 64 px) to attract the dogs' attention to each calibration point. During tests, the dog's trainer was never visible to the dogs. The sample size was determined by the number of chinrest- and calibration-trained dogs available at the time of data collection. The data and code for statistical analyses of this experiment are available at: https://doi.org/10.5281/zenodo.17523757 (Lonardo et al., 2026).

1.1. Stimuli

The familiarisation video showed two actresses throwing a Frisbee back and forth between each other for ten times (total duration of the video: 10 s), a scene that has been found to capture and hold dogs' attention (Völter et al., 2020). The agents were seen from their side and were facing each other, standing (Fig. 1a). This sequence was followed by an attention getter and only when the dogs fixated on it for 200 ms, was one test video presented. We decided to present the same familiarisation video before each test video. The familiarisation introduced the scene and interacting actresses that would have also appeared in the test, but there was no obvious communicative signal yet. We decided on this design because we wanted to test the expectations the dogs bring to the lab due to their previous life experience with humans, rather than building expectations during the experiment with habituation trials (Sirois, 2022).

In test videos, the same two actresses and set-up as in the familiarisation were shown (Fig. 1*a*). The agent on the left side of the screen either gestured for the other agent to approach, for example by patting her hands on her thighs and beckoning (Fig. 1*b*) or to move away, by momentarily pointing in the other agent's direction (Fig. 1*c*). The non-verbal communication took approx. 3 s and was followed by the reaction of the agent on the right side of the screen, which could be either congruent with the communicated gesture (approaching when invited to do so or moving away when invited to do so) or incongruent with it (approaching when invited to move away or moving away when invited to approach). The reaction of the agent on the right side of the screen took approx. 3 s (Fig. 1*d-e*). Hence, we had two congruent conditions (come-approach: CC, go away-go away: GG) and two incongruent conditions (come-go away: CG, go away-approach, GC). The videos were edited so as to show exactly the same footage of the communicative cues between the two conditions that shared the same communicative cues (same left half of the video) and exactly the same footage of the reaction (same right half of the video) between the two conditions that shared the same reaction. This was crucial to ensure that brightness was identical between these conditions and that the same reaction video was supposed to be surprising in one case but not in the other.

1.2. Subjects

We tested 39 dogs (18 females) of different breeds and mixes, mostly mesocephalic (Table S1 reports more demographic and counterbalancing information). Their average age was 59.3 months. Three sessions, from three different dogs, were repeated due to the dogs leaving the chinrest before the reacting agent started to move. Two dogs became unavailable before the end of the data collection (one dog had 3 missing sessions – CC, CG, GC; the other dog had 1 missing session - GC). Nine further dogs had trials in which their gaze was detected in less than 70% of the relevant interest period (see below). To ensure a within-subject comparison, we only retained dogs with a full dataset (all four valid trials, one per condition) in the pupil size analysis ($N = 28$).

1.3. Design

Dogs were tested on one test trial for each of the four conditions, since we were interested in comparing the looking times between the two incongruent and the two congruent trials. On each of four sessions a dog would see the familiarisation video (10 s, including 1 s at the beginning where the video is frozen) and one of the test videos (19 s, including 2 s at the beginning and 10 s at the end where the video is frozen), separated by an attention getter. Consecutive sessions were separated by at least one week.

We counterbalanced across dogs whether the first presented condition shows a congruent or an incongruent action. We pseudo-randomised the order of presentation of the 4 conditions with the constraint that the 4 conditions are equally likely to be presented first

and the two conditions alternated from trial to trial (i.e., CICI or ICIC). We chose this option, instead of a blocked order of conditions, because it might minimise the risk that dogs build expectations about the second agent (e.g., "she always obeys/disobeys"). The design was balanced across dogs with regards to age and sex.

1.4. Analyses

Data were extracted using in DataViewer (version 5.1.1). All analyses were performed in R (R Core Team, 2025; version 4.2.2). The analyses of dwell times and gaze transitions between the agents' AoIs focused on an interest period (IP) spanning the last 13,200 ms of the test videos, from the moment the agent on the right started reacting (approaching or walking away).

1.4.1. Dwell times

We analysed the proportion of time the dogs spent making fixations (EyeLink parsing algorithm) in the agents' AoIs (relative to the time they spent looking at the screen during the whole IP), using GLMMs with beta error structure. We included as sole test predictor the interaction between trial type (congruent vs incongruent) or condition (CC, CG, GC, GG), in two separate models, and agent type (communicating vs reacting) as well as their fixed effects.

Both models included the fixed effect of trial number (z-transformed) as control predictor and the random slope of the test predictor (manually dummy coded and centered) and trial number (z-transformed) within the random intercept of subject ID. We scaled the dependent variables to ensure they did not contain the exact values of 0 and 1 (Smithson & Verkuilen, 2006).

1.4.2. Transitions between agents' AoIs

To analyse the effect of trial type on the number of gaze transitions dogs exhibited between the two agents' AoIs, we used a Generalized LinearMixed Model (GLMM; (Baayen, 2008) with Poisson error structure and loglink function (McCullagh & Nelder, 1989). We included as sole test predictor trial type (congruent vs incongruent) and trial number (z-transformed) as control predictor (both with fixed effect). To account for repeated observations from the same subjects, we included the random intercept of subject ID and the random slope of trial number (z-transformed) and trial type (manually dummy coded and centered) to allow for inter-individual variation in the effect of these predictors on the number of transitions. Moreover, we included in the model an offset term (McCullagh & Nelder, 1989) to control for the duration of looking time (log transformed, base *e*) during each trial, since more transitions between AoIs are possible when dogs look longer to a trial.

1.4.3. Pupil size

We considered as baseline period the 1000 ms of test videos between the end of the signalling actress's gesture and the beginning of the movement of the reacting actress (between seconds 5 and 6 of the videos). Four trials were excluded from these analyses because the dogs' pupil was detected for less than 70% of the whole IP (which lasted 13,900 ms, from the beginning of the baseline until the end of the video). Eight additional trials were excluded because, even though the eye was tracked correctly, the dogs' gaze was detected on-screen for less than 70% of the whole IP, suggesting that dogs were keeping their position on the chinrest but probably not paying attention to the stimuli in those trials. Even when including these 12 trials, dogs' eyes were detected for 95% of the time, averaging across dogs and trials, and their gaze was detected as on-screen for 92% of the time. To ensure a within-subject comparison, we only retained dogs with four valid test trials, one per condition ($N = 28$). Pupil size was analysed during a 4-s IP immediately following the beginning of the reaction (approach or walk away) of the actress who had witnessed the communicative gesture (from s 6 to s 10). We chose the 70% valid data cut-off and the 4-s measuring IP for consistency with our previous eye-tracking studies with dogs (e.g., Lonardo et al., 2024).

The raw pupil size data were pre-processed following standard guidelines in the field (Kret & Sjak-Shie, 2019; Mathôt et al., 2018). Specifically, pupil and gaze samples from the 100 ms preceding and following each blink were removed to decrease the likelihood of artifacts in the signal. We subsequently filled the resulting gaps of up to 500 ms using a linear interpolation. Next, we calculated the median speed of pupil diameter changes for each trial. We excluded samples associated to speed dilation outliers, defined as speed values exceeding eight times the median absolute deviation (MAD; Leys et al., 2013) and repeated the 500-ms linear interpolation. We then subtracted the median pupil size during the baseline period from the interpolated data for each dog and trial. Finally, we downsampled the pupillometry and gaze coordinates data to 10 Hz, by retaining the median value for each time bin, to reduce autocorrelation in the signal, potentially deriving from the relatively slow pupil dilation response compared to the high original (1000 Hz) sampling rate. Finally, we repeated the MAD-based artefact correction and interpolation.

To the pre-processed and down-sampled data, we fitted a generalized additive mixed model (GAMM) with Gaussian error structure, using the bam function of the R package mgcv (version 1.8.41; Wood, 2011), with the smoothing parameter selection method set to "ML" (van Rij et al., 2019). We included a linear term for trial type (congruent or incongruent reaction), our test predictor, and one for session number (1 to 4), a control predictor. We included a smooth term for time and for the different levels of trial type over time (with the maximum number of basis functions set to 20). This allows us to model the different effects of time depending on the level of trial type, since we expected the effect of time on pupil size to be non-linear but not necessarily identical between the levels of trial type. We additionally included a smooth term for the interaction between the X- and Y-gaze coordinates, as the gaze position on the screen might influence pupil size (Mathôt et al., 2018). Finally, we included random factor smooths term for subject and for each recording over time. To evaluate the significance of trial type, we visually inspected the estimated differences between congruent and incongruent trials using the function plot_diff of package itsadug. We assessed autocorrelation in the GAMMs by visually inspecting the correlations between the residuals and the lagged residuals (function acf of the package stats; R Core Team, 2025). We checked the normality and homogeneity of the distribution of the residuals by inspecting a QQ-plot and a plot of residuals against fitted values (function gam.check, package mgcv). These checks confirmed that the autocorrelation was not excessive and that the assumptions of the models were met.

2. Results

2.1. Dwell times

Dogs looked at the agents on average for 4401 (SD ± 3668) ms (CC) and 3967 (SD ± 3296) ms (GG) in the congruent conditions and for 3939 (SD ± 3549) ms (CG) and 4390 (SD ± 3464) ms (GC) in the incongruent ones. A beta-distributed GLMM fitted on the proportion dwell time revealed no significant interaction between trial type and agent type ($\chi^2 = 0.05$, df = 1, $p = .818$). A reduced model without the interaction term, provided no evidence for a significant difference in dwell times between the congruent or incongruent trial type ($\chi^2 = 0.12$, df = 1, $p = .731$; Table S2 and Fig. 2*a*). Dogs looked significantly longer at the reacting than the communicating agent (average reacting agent: 5105 ± 1924 ms, average communicating agent: 3218 ± 2065 ms; fixed effect of agent type: $\chi^2 = 13.85$, df = 1, $p < .001$; Table S2), consistently with the fact that the reacting agent was the only one moving during the IP (Fig. 3).

A second beta GLMM with condition (CC, CG, GC, GG) instead of trial type (congruent, incongruent) as test predictor confirmed that dogs' proportion of dwell time to the agents did not significantly depend on condition either nor on the interaction between condition and agent

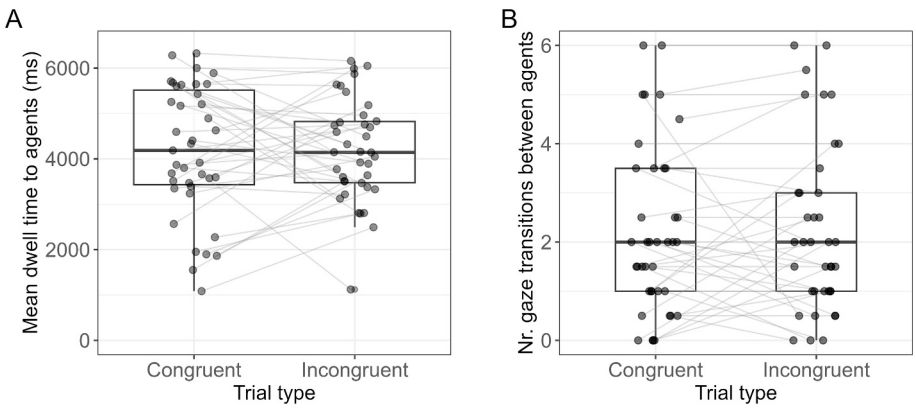


Fig. 2. Boxplots showing dogs' dwell times to the agents (*A*) and their number of gaze transitions between the two agents' AoI (*B*), both as a function of trial type. No statistically significant differences were found between congruent and incongruent trials in either dependent variable. Individual points show average values across trials per subject ($N = 39$) and trial type. The grey lines connect observations belonging to the same subject.

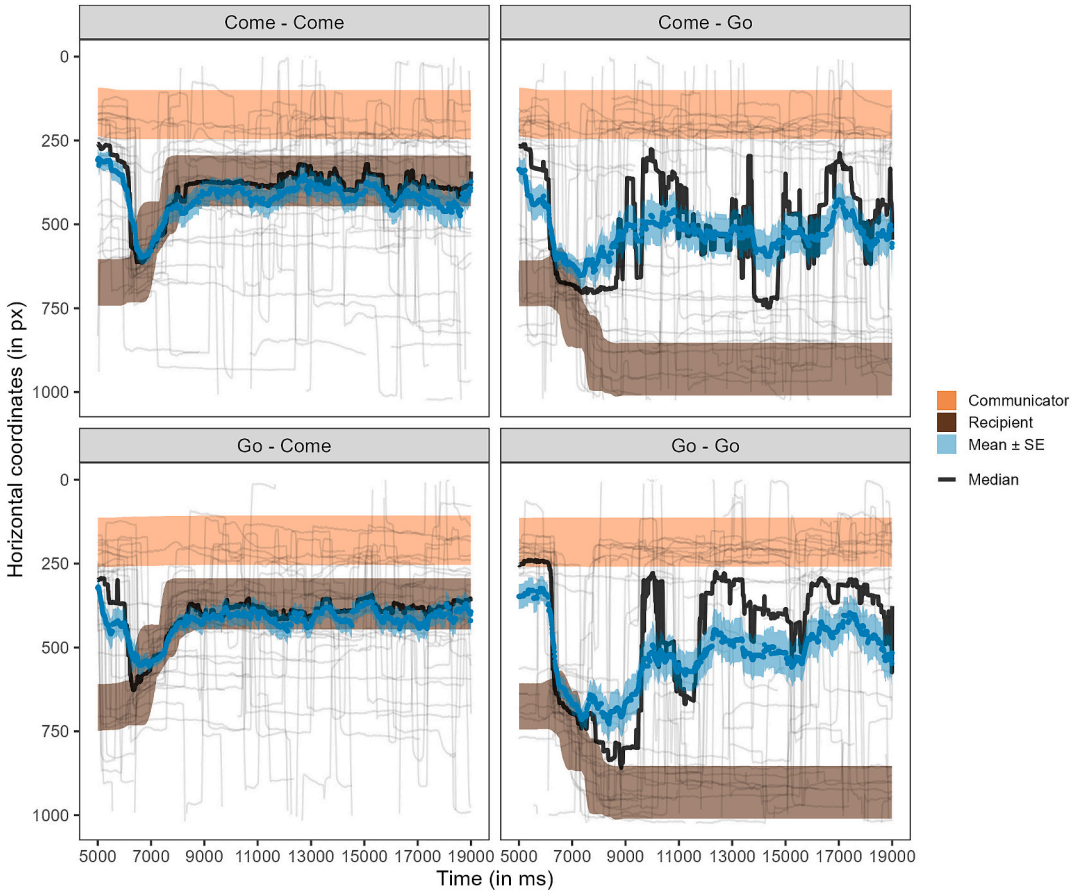


Fig. 3. Time course of the dogs' horizontal gaze coordinates relative to the depicted agents across the four conditions (panel names indicate first the communicating agent's gesture and then the recipient's reaction). The grey lines show the individual on-screen gaze coordinates (dogs with valid trials in all four test trials; $N = 28$). Blue line and the shaded area around it show the dogs mean horizontal gaze coordinates and the standard error of the mean; the black line shows the dogs median horizontal gaze coordinates. The location of the communicating agent (communicator) is shown in orange; the reacting agent (recipient) is shown in brown. The areas of the agents were extracted from the video using the Segment Anything Model 2 (Ravi et al., 2024). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

type (comparison full-reduced model, lacking the fixed effect of condition and its interaction with agent type: $\chi^2 = 10.32$, df = 6, $p = .112$).

2.2. Transitions between agents' AoIs

On average dogs exhibited 2.22 (SD ± 1.98; range 0–8) transitions between the two agents' AoIs per trial. Gaze transitions between the two agents did not differ significantly depending on the trial type ($\chi^2 < 0.01$, df = 1, $p = .951$; Table S3 and Fig. 2*b*). With increasing session number, dogs tended to exhibit less gaze transitions between the two agents ($\chi^2 = 2.95$, df = 1, $p = .086$).

2.3. Pupil size

Dogs' pupil size was significantly larger when dogs witnessed the incongruent than the congruent reactions to the communicative gestures (linear effect of trial type: $t = 2.04$, $p = .041$). In particular, this difference between incongruent and congruent trials was significant between ms 7697 and ms 9111 (red line in Fig. 4B), i.e., approx. 1.8 s after the reacting actress started to move. A Wilcoxon signed-rank test of the dogs' mean pupil size in the same interest period confirmed that incongruent trials elicited larger pupil sizes than congruent trials ($T^+ = 313$, $N = 28$, $p = .011$). Session number did not affect pupil size linearly ($t = -0.01$, $p = .990$). The non-parametric effect of time ($F = 11.87$, $p < .001$), gaze coordinates on the screen ($F = 31.14$, $p < .001$), subject over time ($F = 0.09$, $p < .001$) and the time series of each recording ($F = 90.57$, $p < .001$) significantly contributed to explaining the pupil size. The results of the GAMM analysing pupil size ($N = 28$) are shown in Table S4 and its partial effects are plotted in Fig. S2. When retaining all data for the analyses ($N = 39$), i.e., including the 11 dogs who did not provide valid data in each of the 4 test trials and including the trials in which dogs' gaze was detected and on-screen for less than 70% of the time, the parametric effect of trial type was not significant ($t = 0.54$, $p = .592$). The non-parametric effect of time, gaze coordinates on the screen, and the combination of subject, condition and session number still significantly contributed to explaining the pupil size (Table S5; Fig. S3 for partial effects plot).

3. Discussion

The aim of this eye-tracking study was to probe dogs' expectations about human social conventions, using eye tracking and pupillometry. Dogs' pupil size changes revealed their sensitivity to human conventional gestures, specifically to the typical reaction to some human communicative gestures. We found this difference even though dogs' dwell times and number of gaze alternations between the two agents did not differ significantly.

Dogs paid close attention to the stimuli. On average, their gaze was detected by the eye-tracker and on-screen for 92% of the time and they alternated their gaze between the two agents' AoIs, suggesting that they recognized the socially relevant parts of the scenes, consistently with the findings of a previous eye-tracking study with dogs (Törnqvist et al., 2020). In particular, their vertical gaze coordinates (Fig. S1) suggest that dogs focused their attention on the agents' faces and their dwell times as well as their horizontal gaze coordinates (Fig. 3) indicate that they closely followed the reacting agent's movements.

Importantly, our design can rule out that dogs' pupillary response is due to low-level (perceptual) features of the stimuli (including luminosity) because each video showing the reaction of the (approaching or walking away) agent served, in one condition, as the surprising (incongruent) stimulus and, in the other condition, as the expected (congruent) stimulus relative to the previous communicative gesture. This finding suggests that dogs might have third-party expectations about the appropriate response to human conventional gestural communication. Further, the pupillometry finding suggests that these expectations about what the agent will do are context-specific: the same action is considered in one case surprising and in the other case expected. There is evidence of animals (mainly apes) being sensitive to conventions in relation to the self, yet so far expectations about conventions that do not directly involve the observer have not been reported in non-human animals (Schmidt & Rakoczy, 2023). When addressed directly and ostensively, dogs are skilled at following human deictic gestures to locate hidden food (Kaminski & Nitzschner, 2013; ManyDogs Project, et al., 2023). In the present study, we found indirect evidence that, in the context of a third-party interaction between two humans, dogs expected a pointing gesture away from the communicator and in the direction of the recipient of the signal, to lead the recipient to move in the signalled direction. They also expected that the communicator tapping with her hands on her thighs and beckoning would lead to the recipient to move closer. Dogs' pupil sizes reacted to the incongruent actions despite the dogs never being addressed by the depicted agents. Hence, dogs' expectations in this study were not self-centred and extended to third-party interactions between two humans.

Conventions, defined as behavioural regularities exhibited by a majority in recurring social circumstances (Lewis, 1969), allow observers to form expectations about how other individuals *will* typically behave in such circumstances. Importantly, however, conventions are different from norms in that they do not inform about how individuals *ought to* behave in a given context (Bicchieri, 2006). We argue that the present findings highlight dogs' sensitivity to human social conventions, i.e., dogs' expectations about the typical reaction an individual will show, but more research is needed to assess whether dogs might also understand social norms, intended as an understanding of how an individual ought to behave in a certain social situation. Interestingly, however, scholars approaching norm psychology from a comparative perspective have criticized heavily mentalistic accounts of norm psychology, such as the one just described, and extend at least simpler notions of social norms also to non-human species (Andrews et al., 2024;

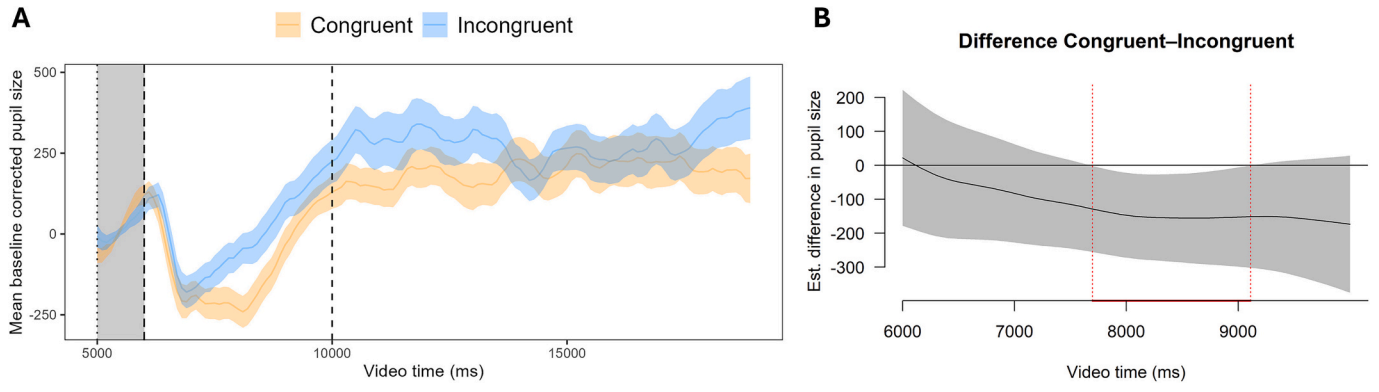


Fig. 4. Pupillometry time course. A) Time series plot showing dogs' pupil size (in arbitrary units and baseline corrected). The orange and blue lines show the mean pupil size (±se, coloured shaded area around the lines) in the Congruent and Incongruent trial types, respectively. The grey shaded area at the beginning of the plot shows the 1000 ms that we used as baseline (between seconds 5 and 6 of the plot). The measuring IP (from second 6 to 10) is shown between dashed vertical lines. B) The estimated difference in pupil size between trial types (Congruent - Incongruent) is shown during the measuring IP. The grey shaded area shows the 95% confidence interval. The dogs' pupil size is significantly larger in the Incongruent than Congruent trial types (highlighted in red) between ms 7697 and ms 9111(the reacting agent starts to move around ms 6000). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Westra et al., 2024). In this light, a few psychological processes are likely to give rise to normative regularities across species, and the present findings could indicate that dogs have at least one such process: a "submentalistic behaviour understanding" (Andrews et al., 2024), i.e., in this case, dogs formed expectations about the future reaction of the receiver of communicative gestures based on the dogs' own previous learning experience (associative learning through direct reinforcement or observational learning). Associative and reinforcement learning are also considered building blocks of human norm psychology (Andrews et al., 2024). Based on the present findings, we do not claim that dogs in their own behaviour follow human-like conventions nor that their expectations are necessarily homologous to those that humans might form watching the same stimuli. What our findings suggest is that dogs can detect violations of behavioural regularities when observing incongruent human social interactions and thus dogs' expectations about some human communicative conventions might be functionally similar to those of humans.

Showing non-conspecific actors might make processing social interactions more demanding for dogs (Törnqvist et al., 2015). However, we deliberately chose to show videos of human agents to investigate if domestic animals with extensive exposure to human social conventions would come to have expectations about some human-typical conventions, and the pupillometry results provide initial evidence in favour of this hypothesis.

The dwell times confirmed that dogs paid attention to the human agents producing and reacting to human-typical communicative gestures, even if the agents never addressed the dogs directly. Yet, dogs looked at the reacting actress equally long irrespective of whether her reaction was congruent or incongruent with the communicative gesture she had previously received.

Dogs' gaze pattern suggests that they were following the actions of the two humans in all videos to a similar extent. Indeed, on average they looked for about 4.4. s at the approaching actress, (irrespective of whether her approach was preceded by an invite to approach or by a signal to move away from the signalling actress) and for about 3.9/4.0 s at the agent moving away (irrespective of whether this was preceded by an invite or a signal to move away). This suggests that strictly matching the stimuli across conditions and controlling for low-level visual features of the stimuli ensuring comparable visual engagement across conditions is an important methodological concern for pupillometry studies with dogs.

One might wonder why dogs' dwell times and gaze alternations did not significantly differentiate between congruent and incongruent responses, while their pupil size changes did. First, the dogs' chin-rest training may have led to a ceiling effect in their dwell time data. Similar to adult human participants in eye-tracking studies, most of the dogs in our study stayed on the chin-rest and looked on the screen throughout the test trials, which might have reduced the variance in dwell times we might have observed otherwise. Second, eye movements (including measures such as dwell times and AoI transitions) may be less sensitive measures of expectancy violations than pupil dilation. This hypothesis is consistent with findings from developmental psychology with human infants and toddlers (Geangu et al., 2011; Hepach & Westermann, 2013; Sirois, 2009), as well as with a previous study with dogs (Lonardo et al., 2024). However, given the lack of effects on dogs' looking patterns, the pupillometry results need to be replicated and dogs' understanding of social expectations warrants further investigation, especially given that in this study we only probed dogs' sensitivity to the social conventions behind two communicative gestures.

Our study does not reveal *when* and *how* dogs acquire their expectations regarding the common reactions to human conventional gestures. For example, these expectations might originate from explicit training or mere exposure to human communicative gestures (observational learning). Either of these learning mechanism could result in expectations concerning the likely (i.e., more frequently co-occurring) reactions to the gestures presented in our stimuli. Our study shows, however, that these expectations extend to third-party interactions between two humans presented on a screen. Therefore, the dogs either acquired these expectations through first-person interactions with humans and transferred them to a third-person (human-human) context, or they acquired these expectations by observing human or dog-human interactions. Both possibilities would highlight how flexibly dogs keep track of the communicative interactions in their social environment and thus our findings pave the way for further investigations into dogs' sensitivity to human social conventions and its origins.

CRediT authorship contribution statement

Lucrezia Lonardo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Christoph J. Völter:** Conceptualization, Formal analysis, Methodology, Software, Validation, Visualization, Writing – review & editing. **Ludwig Huber:** Conceptualization, Funding acquisition, Methodology, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supplementary material

Supplementary material can be found online at https://doi.org/10.1016/j.cognition.2026.106586.

Data availability

The data and code for statistical analyses of this experiment are available at: https://doi.org/10.5281/zenodo.17523757

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